

## Mark schemes

### Q1.

|      | Marking Instructions                                                                                                   | AO     | Marks | Typical Solution                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------|--------|-------|-------------------------------------------------------------------------------------|
| (a)  | Uses correct formula for impulse with one velocity negative                                                            | AO1.1a | M1    | $I = mv - mu$<br>$I = 250(1.2) - 250(-1.8)$                                         |
|      | Obtains correct answer of 750                                                                                          | AO1.1b | A1    | $I = 750$                                                                           |
| (b)  | Forms an equation involving appropriate integral using 'their' value from part (a) to find $k$                         | AO3.4  | M1    | $750 = \int_0^{0.8} k(4 - 5t) dt$<br>As $\int_0^{0.8} t(4 - 5t) dt = \frac{32}{75}$ |
|      | Evaluates definite integral correctly                                                                                  | AO1.1b | B1    | Then $k = \frac{750}{\frac{32}{75}} = \frac{28125}{16}$                             |
|      | Solves equation to find the value for $k$ – follow through their '750'                                                 | AO1.1b | A1F   | $= 1800$ (2sf)                                                                      |
|      |                                                                                                                        |        |       |                                                                                     |
| (ii) | Deduces when the maximum value of the force occurs and substitutes correct value of $t$ into the formula for the force | AO2.2a | M1    | Maximum occurs at $t = 0.4$<br>Maximum value is<br>$v_B = \frac{u(3 - 2c)}{5}$      |
|      | Obtains correct value for 'their' maximum force – follow through incorrect $k$                                         | AO1.1b | A1F   | $= 1400$ (2sf)                                                                      |
|      |                                                                                                                        |        |       | <b>Total 7 marks</b>                                                                |

### Q2.

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| Marking Instructions   | AO     | Marks | Typical Solution                                       |
|------------------------|--------|-------|--------------------------------------------------------|
| Circles correct answer | AO1.1b | B1    | $I = \frac{1}{2} \times 0.8 \times 4 = 1.6 \text{ Ns}$ |
| <b>Total 1 mark</b>    |        |       |                                                        |

### Q3.

|     | Marking Instructions                   | AO     | Marks | Typical Solution                        |
|-----|----------------------------------------|--------|-------|-----------------------------------------|
| (a) | Deduces correct value for $b$ .        | AO2.2a | B1    | $b = 0.02$                              |
| (b) | Forms an integral to find the impulse. | AO3.4  | M1    | $I = \int_0^{0.02} kt^2(t - 0.02)^2 dt$ |
|     | Integrates terms and uses              | AO1.1a | M1    |                                         |

|     |                                                                                              |        |     |                                                                                                                                |
|-----|----------------------------------------------------------------------------------------------|--------|-----|--------------------------------------------------------------------------------------------------------------------------------|
|     | limits or uses a calculator for definite integral (PI)                                       |        |     | $= k \int_0^{0.02} (t^4 - 0.04t^3 + 0.0004t^2) dt$                                                                             |
|     | Obtains correct value for impulse. (AWRT $1.1 \times 10^{-10}$ )                             | AO1.1b | A1  | $= k \left[ \frac{t^5}{5} - \frac{t^4}{100} + \frac{t^3}{7500} \right]_0^{0.02}$ $= k \times 1.07 \times 10^{-10} \text{ N s}$ |
| (c) | Uses 'impulse equals change in momentum' to form an equation, with 'their' impulse from (a). | AO3.4  | M1  | $k \times 1.07 \times 10^{-10} =  0.15 \times 4 - 0.15 \times (-2) $                                                           |
|     | Obtains a correct equation for 'their' impulse.                                              | AO1.1b | A1F | $k = \frac{0.9}{1.07 \times 10^{-10}}$                                                                                         |
|     | Obtains the correct value for $k$ .<br>CAO                                                   | AO1.1b | A1  | $= 8.4 \times 10^9$                                                                                                            |
|     |                                                                                              |        |     | <b>Total 7 marks</b>                                                                                                           |

**Q4.**

Use of Impulse-momentum principle

$$\int_{(0)}^{(r)} (3t + 1) dt = \pm 2(5) \pm 2(1)$$

Condone sign error for M1

M1

$$\int_{(0)}^{(r)} (3t + 1) dt = 2(5) - 2(1)$$

A1 for all correct

A1

$$\left[ \frac{3}{2}t^2 + t \right]_{(0)}^{(r)} = (8)$$

Correct integration, PI by the correct quadratic

A1

$$3T^2 + 2T - 16 = 0$$

Correct use of correct limits and rearrangement

A1

$$\left. \begin{aligned} (3T+8)(T-2) &= 0 \\ \text{or } T &= \frac{-2 \pm \sqrt{4 - 4(3)(-16)}}{2(3)} \end{aligned} \right\}$$

Solution of their quadratic, **correct** attempt needed

m1

$$\begin{aligned} (T = -\frac{8}{3} \text{ unacceptable, not} \\ \text{in the interval } 0 \leq t \leq 3) \\ \underline{T = 2} \end{aligned}$$

A1

[6]

Q5.

$$(a) \quad I = \int_0^{0.5} 4 \times 10^4 t^2 (1 - 2t) dt$$

Attempt to integrate

M1

Use of correct limits, PI

A1

$$= 4 \times 10^4 \left[ \frac{1}{3} t^3 - \frac{1}{2} t^4 \right]_0^{0.5}$$

Correct integration

A1F

$$= 417 \left( \text{or } \frac{1250}{3} \right) \text{ Ns}$$

Accept 416.6 or 416.7

A1F

4

$$(b) \quad 416.6 = 60V + 60 \times 5$$

A1F correct sign

M1A1F

$$V = 1.94$$

AWRT 1.94, accept 1.95  
ISW

A1F

3

[7]

**Q6.**

(a)  $I = 0.2(32) + 0.2(18)$

M1

$I = 10 \text{ N s}$

*Condone +10*

A1

2

(b)  $\int_0^{0.09} k(0.9t - 10t^2) dt = 10$

*Condone limits*

M1

$k \left[ 0.45t^2 - \frac{10}{3}t^3 \right]_0^{0.09} = 10$

*Condone limits*

A1F

$1.215 \times 10^{-3}k = 10$

*For substituting 0.09*

m1

$k = 8230$

A1F

4

[6]

**Q7.**

(a)  $I = \int_0^4 (t^3 + t) dt$

M1

$= \left[ \frac{1}{4}t^4 + \frac{1}{2}t^2 \right]_0^4$

m1

$= 72 \text{ N s}$

A1

3

(b)  $72 = 0.5v - 0.5(0)$

*Condone - 5(0)*

M1

$v = 144$

A1F

2

(c) 
$$\int_0^T (t^3 + t) dt = 0.5(12) - 0.5(0)$$

*Condone – 5(0)*

M1

$$\left[ \frac{1}{4}t^4 + \frac{1}{2}t^2 \right]_0^T = 6$$

$$T^4 + 2T^2 - 24 = 0$$

A1

$$T^2 = \frac{-2 \pm \sqrt{2^2 - 4(1)(-24)}}{2(1)}$$

m1A1F

or  $(T^2 - 4)(T^2 + 6) = 0$

$$T^2 = 4$$

$$T = 2$$

A1F

5

[10]

**Q8.**

$$\int_0^t 5 \times 10^3 t^2 dt = 0.2(2) - 0.2(0)$$

*Impulse-Momentum principle*

M1A1

$$\frac{5 \times 10^3}{3} t^3 = 0.4$$

A1F

$$t = 0.0621$$

*At least 3 sig. fig. required*

A1F

[4]

**Q9.**

(a)  $I = \int_0^3 (4t + 5) dt$

M1

$$= \left[ 2t^2 + 5t \right]_0^3$$

*Or evaluation of constant*

m1

$$= 33 \text{ Ns}$$

A1

3

**Alternative:**

$I = \text{Area under the Force–Time graph}$

$$= \frac{17 + 5}{2} \times 3$$

(M1)

$$= 33 \text{ Ns}$$

(m1)

(A1)

(3)

(b)  $I = mv - mu$   
 $33 = 2v - 2(0)$

M1

$v = 16.5 \text{ m s}^{-1}$

A1F

2

(c)  $I = \int_0^t (4t + 5) dt = 2(37.5) - 2(0)$

M1

$$2t^2 + 5t - 75 = 0$$

A1

$$t = \frac{-5 \pm \sqrt{25 + 8 \times 75}}{4}$$

m1

$t = 5$

For one value of  $t$  identified only

A1F

4

[9]

**Q10.**

(a) 
$$I = 1.4 \times 10^5 \int_0^{0.1} (t^3 - 10t^4) dt$$

M1A1

$$= 1.4 \times 10^5 \left[ \frac{1}{3} t^3 - \frac{10}{4} t^4 \right]_0^{0.1}$$

$= 11.7 \text{ Ns}$

AG

m1

A1

4

(b) initial momentum =  $0.45(-15)$

M1

$= -6.75 \text{ Ns}$

final momentum =  $11.7 - 6.75$

M1

$= 4.95 \text{ Ns}$

velocity after impact  $= \frac{4.95}{0.45}$

dependent on both previous M1s

m1

$= 11 \text{ ms}^{-1}$

A1

4

- (c) The ball is not perfectly elastic  
or  $e \neq 1$  or energy loss

E1

1

[9]

**Q11.**

(a)



$$\text{impulse} = mv - mu$$

*attempt to use  $|mv - mu|$*

M1

$$= 240(1) - 240(-2)$$

*correct signs*

A1

$$= 720 \text{ N s}$$

*must have units; ft applies to 240 N s only*

A1ft

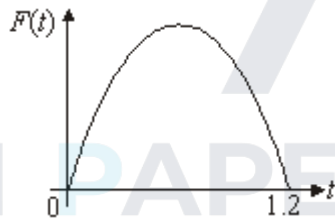
3

(b) (i)  $t = 0, 1.2$

B1

1

(ii)



*shape*

B1

*symmetrical / axis / labels*

B1

2

(iii) max when  $t = 0.6$  (symmetry)

*t value found/stated*

B1

$$F(0.6) = 500(0.6)(6 - 5 \times 0.6)$$

*attempt to find F*

M1

$$= 300 \times 3$$

$$= 900 \text{ N}$$



AG

A1

3

(iv) total area below curve = impulse magnitude

E1

1

[10]

**Q12.**

(a)  $A \xrightarrow{43.2}$   $B \xrightarrow{0}$   
 $1200 \xrightarrow{v}$   $1500 \xrightarrow{u}$

Conservation of momentum

$$43.2(1200) = 1200v + 1500u$$

*Use of  $\text{km h}^{-1}$  or  $\text{ms}^{-1}$  throughout*

M1A1

$$43.2 = v + 1.25u$$

(1)

Restitution Law

$$u - v = \frac{1}{4}(43.2 - 0)$$

*M0 if 'e' on the wrong side*

M1A1

$$10.8 = u - v$$

(2)

$$(1) + (2) \quad 54 = 2.25u$$

*Solving a part of equations*

M1

$$u = 24 \text{ km h}^{-1}$$

$$\text{or } 6\frac{2}{3} \text{ ms}^{-1}$$

A1

$$v = 13.2 \text{ km h}^{-1}$$

*Follow through error through error in their*



equations or  $3\frac{2}{3} \text{ ms}^{-1}$

B1f.t.

7

$$(b) \quad 24 \text{ km h}^{-1} = \frac{24 \times 1000}{60 \times 60}$$

$$= 6\frac{2}{3} \text{ ms}^{-1}$$

Conversion of units correct in this part

B1

$$\text{Impulse} = 1500 \left( 6\frac{2}{3} \right) - 0$$

Impulse = change in momentum for the particle

M1

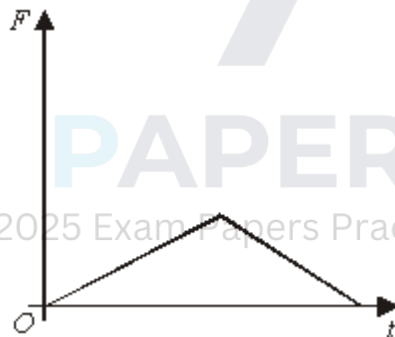
$$= 10\,000 \text{ N s}$$

Printed answer

A1

3

(c) (i)



B1 increase then decrease

B1 all correct

B(2,1,0)

2

$$(ii) \quad \text{Area} \Rightarrow \frac{1}{2} (0.01)F = 10000$$

Use of area – uses printed answer

M1A1

$$F = 2 \times 10^6 \text{ N}$$

Follow through error in area.

A1f.t.

- (iii) Likely to be non-linear, or force wouldn't change abruptly  
*Suitable comment – dependent on graph*

E1

1

[16]

**Q13.**

(a) Using Impulse =  $\int F dt$   
 $= \int_0^5 (3t^2 + 7e^{-t}) dt$

M1

$$= \left[ t^3 - 7e^{-t} \right]_0^5$$

A1

$$= 125 - 7e^{-5} + 7$$

A1

3

$$= 132 - 7e^{-5}$$

- (b) Impulse = change in momentum;

M1

$$132 - 7e^{-5} = 4u - 4.6$$

$$= 4u - 24$$

$$u = 39 - \frac{7}{4}e^{-5}$$

*Accept 38.988*

A1

2

[5]

**Q14.**

Gain in momentum =  $\int_0^4 F dt$

M1

$$= \left[ -3e^{-2t} \right]_0^4$$

A1 M1

$$= 3 - 3e^{-8}$$

A1

[4]

Q15.

(a)  $I = \int F dt$

M1

$$= \int_0^{0.2} 30t(0.2 - t) dt$$

$$= [3t^2 - 10t^3]_0^{0.2}$$

M1A1

$$= 0.12 - 0.08$$

$$= 0.04$$

A1

4

(b) Using impulse = change in momentum  $0.04 = 0.005(6 + v)$

M1 A1ft

for 0.005

B1

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$$= 0.03 + 0.005v$$

Speed is  $2 \text{ ms}^{-1}$

ft dep on M2 in (a)

A1ft

4

[8]

## Examiner reports

### Q1.

In this question, students appeared to have a better understanding of variable impulse than constant impulse. The mean mark for part (a) was 51%, the mean mark for (b)(i) was 69% and the mean mark for (b)(ii) was 36%.

Roughly half of all students obtained full marks for part (a) with the most common issue being a sign error. Students who did not appreciate the need to use opposite signs to calculate the required impulse scored zero marks. It was common to see  $250(1.2) - 250(1.8) = -150$ , which scored zero marks.

Any error in part (a) was followed through, and many students did score full marks on the remaining parts. Part (b) was the most successfully answered, with students using their

calculators to obtain a value of  $\frac{32}{75}$  for  $\int_0^{0.8} t(4 - 5t) dt$  and hence find their value for  $k$  by using their answer from part (a).

Part (c) required students to appreciate that the model for the force was based on a quadratic function and hence to deduce that the maximum value occurred at  $t = 0.4$ . Many students left this part unanswered.

### Q4.

This question was answered fairly well by most students. Most students were able to use the impulse-momentum principle correctly. Some students found the acceleration and solved the differential equation to find velocity as a function of time and hence found the value of  $T$ . The great majority of the students were able to deal with the impulse of the variable force and only a small minority attempted to use the constant acceleration formulae here.

### Q5.

Most candidates found this question straightforward. However, some candidates made sign errors in writing the change of momentum of the ice-hockey player and using the impulse-momentum principle in answering part (b).

### Q6.

The candidates' responses to this question were generally good. However, for part (a), in failing to recognise momentum as a vector quantity, some candidates did not take into account the reversal of the direction of motion of the ball after hitting the bat. Some candidates did not understand how to find the impulse of a variable force. These candidates attempted to answer part (b) by treating the force as constant. There were some candidates who misread 0.09 seconds and instead used 0.9 seconds for the upper limit of their integral. A small number of candidates attempted to integrate the force but actually differentiated it. Almost all candidates showed understanding of the equivalence of impulse and change of momentum.

### Q7.

The candidates' responses to this question were generally very good. However, there was a small number of candidates who did not understand how to find the impulse of a variable force. These candidates attempted to answer parts (a) and (c) by treating the

force  $(t^3 + t)$  N as constant. Almost all the candidates showed understanding of the equivalence of impulse and change of momentum in answering part (b).

Most candidates who answered part (c) correctly found the impulse of the variable force and used the impulse/momentum principle to arrive at the quadratic equation  $t^4 + 2t^2 - 24 = 0$ . They then used the formula to solve the equation to find the time taken by the particle to reach a speed of  $12 \text{ m s}^{-1}$ . Some candidates attempted to solve this equation by taking the square root of each term separately in a bid to change the equation into a quadratic in  $t$ .

### Q8.

This question aimed to test the candidates' understanding of the Impulse/Momentum principle, where the force is a function of time. Many candidates answered this question

correctly. However, some candidates evaluated the integral  $\int_0^{0.1} 5 \times 10^3 t^2 dt$  without having any need for it.

A large number of candidates lost one accuracy mark because they gave the value of  $t$  to less than three significant figures. There were candidates who attempted to use the constant acceleration formulae to find the acceleration, in a bid to find the required value of  $t$ . Some other candidates treated the force as constant and used  $Ft$  for the magnitude of the impulse.

### Q9.

A significant number of candidates did not answer this question correctly. These candidates did not know how to find the impulse of a variable force and they attempted to treat the force as constant.

A majority of those candidates who answered part (a) correctly used integration with appropriate limits to find the impulse. The others used the area under the force–time graph.

It was rather surprising to note that some candidates who answered parts (a) and (b) correctly attempted to answer part (c) by treating the force as constant.

### Q10.

Part (a) A large number of candidates substituted 0.1 for  $t$  in  $1.4 \times 10^5 (t^2 - 10t^3)$ , instead of integrating to find the magnitude of the impulse exerted by the bat on the ball.

In part (b) many candidates could not use the principle of conservation of linear momentum correctly and accurately. These candidates overlooked the change in the direction of the ball due to the rebound.

Part (c) Most candidates stated that the ball is not perfectly elastic, or  $e < 1$ , or some of the kinetic energy of the ball is transferred into heat and sound energies.

### Q11.

There was a very good response to this question. When errors were made they tended to be one of the following: not using different signs for the velocities and therefore getting 240 Ns; failing to realise that the force was a symmetrical curve; not labelling the axes or failing to show 1.2 on the  $t$  axis; or not realising that area = impulse in part (iv). In part (c)

many candidates used calculus methods correctly. Equally, using the fact that the curve was symmetrical with  $t = 0.6$  substituted to get 900N scored full marks.

### Q12.

The required equations were usually set up well. A mark was sometimes lost in part (a) because of a numerical slip. Candidates who used the answer given to find the speed of  $A$  were penalised for not solving a set of simultaneous equations. Candidates who were not familiar with the correct units could not fully complete part (b). Candidates who found the impulse of  $B$  on  $A$  had to make clear reference to “equal and opposite impulses” to gain the final mark for part (b).

In part (c)(i) a few candidates thought that the graph had to be curved and not composed of two straight line segments. Part (c)(ii) proved to be a test of remembering a formula for the area of a triangle more than anything else. The quality of the criticism in part (c)(iii) showed a distinct improvement over previous papers, with candidates often indicating a sound understanding of the situation.

### Q13.

In part (a), a significant number of candidates did not use  $\int F dt$  to find the impulse. A common error of those who did was to find the integral to be  $125 - 7e^{-5} - 7$ , rather than  $125 - 7e^5 + 7$ . Part (b) was answered well.

### Q14.

Those who used the formula “Change in momentum =  $\int F dt$ ” found the correct result quickly.

### Q15.

This question was well attempted. In part (a), a few did not use Impulse =  $\int F dt$ , and some differentiated  $F$ , even though they had written the integral sign. In part (b) the signs of the velocities caused problems with a few not using 5g to be 0.005kg.